

Faris Doughan

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EDUCATION

Harvard University

Bachelor of Arts in Mathematics and Statistics | GPA: 3.7/4.0

Cambridge, MA

Expected May 2028

Relevant Coursework: Studies in Real & Complex Analysis (Math 55); Measure Theory; Probability I & II (Graduate); Statistical Inference; Stochastic Analysis and Applications (Graduate); Random High-Dimensional Optimization (Graduate); Algebraic Fundamentals of Data (Graduate); Advanced Scientific Computing–Numerical Methods (Graduate).

Intend to pursue the concurrent S.M. program in Applied Mathematics.

EXPERIENCE

Undergraduate Researcher

Harvard School of Engineering & Applied Sciences

Feb 2026 – Present

Cambridge, MA

- Summer 2026 (ongoing): Work on generalizing diffusion models—probabilistic generative models that learn data distributions by reversing a gradual noise process. Exploring theoretical extensions of the diffusion framework together with implementations and empirical evaluation. Advised by Dr. John Vastola and Prof. Sam Gershman.

Teaching Assistant

Harvard Data Science Initiative

Mar 2026 – Present

Cambridge, MA

- TA for a generative AI workshop series for PhD students and postdocs, delivering lectures to 75+ attendees covering LLMs, Huggingface, embeddings, RAG systems, agents, and AI ethics.
- Support instruction by debugging Python/Colab/API workflows, and developing weekly exercises for participants.

Course Assistant

Harvard University

Jan 2026 – Present

Cambridge, MA

- Support student learning in lectures, hold office hours, grade problem sets, and provide general instructional support for Math 154: Probability Theory (Spring 2026) and Math 136: Differential Geometry (Fall 2026).

AI & Advanced Data Analytics Intern

Arab Bank

Jun 2025 – Aug 2025

Amman, Jordan

- Boosted production ML model precision from 80% to 94% (+10.5% improvement from cohort) through feature engineering.
- Developed a Power BI and Matplotlib dashboard aggregating model outputs and 20+ KPIs across international branches

Technology Assistant & Digital Reserves Assistant

Harvard Library Assistive Technology Center

Oct 2024 – Dec 2025

Cambridge, MA

- Digitized material and created 125+ accessible PDFs, utilizing Adobe OCR, Scantailor, Adobe Acrobat, and Mathpix.
- Accelerated remediation of math textbooks by over 250% by training a custom ChatGPT-based model to convert table images into properly formatted LaTeX.

EXTRACURRICULAR & OTHER INVOLVEMENTS

Harvard Mathematics Department Directed Reading Program

Reading in Random Matrix Theory (Spring 2026)

Jan 2026 – Present

Harvard Student Agencies

SAT & STEM Tutor for High School Students

Nov 2025 – Present

Other Involvements: Harvard Shooting Club Trap & Skeet Team, Harvard Poker Club, Harvard Open Data Project, Harvard Undergraduate Mathematics Association, Society of Arab Students, Group for Undergraduates in Statistics, Arrowstreet Capital Quant Symposium Invitee

TALKS

Distribution and Asymptotic Limit Behavior in the Robinson–Schensted Correspondence

Math 155R: Algebraic Combinatorics

April 2026

Cambridge, MA

- Delivered a 30-minute talk on the combinatorial algorithm for random permutations and its distribution.

Random Walks on Simplicial Complexes and Expansion Properties

CS 2253: Graduate Seminar on High-Dimensional Expanders

March 2026

Cambridge, MA

- Delivered a 75-minute seminar lecture explaining how random walks on higher-dimensional structures extend classical graph concepts and their relevance in modern theoretical computer science.

SKILLS

Technical: Python, Java, JavaScript, R, SQL, HTML/CSS, Jupyter, Tableau, Power BI

Languages: English (Native), Arabic (Native), Spanish (Proficient)